LECTURE 05 | START HERE

Prototyping and interface walkthroughs

The missing lecture, organized for understanding, concise recall and application.

This fills the previous gap

The website now includes **all six supplied lectures**. Lecture 5 provides Andriy's own material on heuristic evaluation, cognitive walkthroughs, GOMS, KLM and Fitts' Law. The other professor's exam photos remain a separate supplemental source.

Pages	What you will find
2	Conceptual models, metaphors and interaction styles
3-5	Fidelity, coverage, lifecycle and prototyping techniques
6-7	Design principles, simplicity, learnability and visual design
8-10	Participatory design, heuristic evaluation and cognitive walkthroughs
11-12	GOMS, KLM and Fitts calculations; final revision checklist
13-15	24 short-answer questions with two-line models
16-20	40 original MCQs in website source order
21-22	MCQ answer key, explanations and exact source-page references

Five distinctions to learn first

Conceptual model / styling: organization and relationships / visual presentation.
Fidelity / coverage: realism of representation / supported features and paths.
Horizontal / vertical: broad surface / selected depth.
Heuristic evaluation / walkthrough: rule-based inspection / new-user task discovery.
KLM / Fitts: estimated time for an action sequence / time for aimed pointing.

A useful answer pattern

For a comparison, state one side per line. For a design problem, name the concept and evidence, then give a targeted fix. For calculations, show the formula, operator counts or ratio before the result. Do not present a model estimate as measured user performance.

Examples and practice questions are original study aids. They do not predict an exam or supply an official marking scheme.

CONCEPTUAL DESIGN

Explain how it works before styling it

A **conceptual model** is a high-level description of organization, operations and relationships. It gives the team orientation, shared terms and questions to investigate while keeping alternatives open. It should expose assumptions rather than decorate an untested idea.

Nutrition example: users record foods and quantities into meals; meals belong to days; corrections update the daily summary. That relationship is part of the model. Green buttons and a particular typeface are visual choices, not the model itself.

Metaphors transfer knowledge - selectively

Calendars, shopping carts, desktops and folders borrow familiar knowledge. A metaphor helps explain unfamiliar behaviour, but copying every physical property may limit software unnecessarily. Software can support undo or resized windows even where the physical analogy does not.

Interaction type	Core idea	Example
Instructing	Issue commands or select options.	Choose Save; issue a voice command.
Conversing	Interact as if in a conversation.	Ask a question and respond to clarification.
Manipulating	Act on represented physical/virtual objects.	Drag an item into a folder; use a two-finger gesture.
Exploring	Move through virtual environments or physical spaces.	Navigate a virtual museum.

Interface styles are choices within a design

Command language: powerful for experienced users, but unfamiliar syntax raises learning/recall demands.
Menus/forms: present recognizable choices; labels and structure still matter.
Speech dialog: can suit novice or infrequent users, but people may not know what requests are supported. Natural language is not a guarantee of discoverability.

Direct manipulation: all three defining features

1. Continuous representation of relevant objects and actions.
2. Physical actions/button pressing rather than complex command syntax.
3. Rapid, reversible actions with immediate feedback on the object of interest.

A screen full of icons is not automatically direct manipulation. Check what users can act on, what happens immediately, and whether the action is reversible. Some operations are awkward to express physically or need too much screen space.

PROTOTYPE FIDELITY

Build enough realism to answer the question

Why prototype?

Explore competing ideas with low investment. Make them concrete enough for users to react, suggest changes and reveal mismatches. Rapid revision matters: if the team is too invested to discard the prototype, it becomes harder to investigate alternatives honestly.

Representation	Useful for	Limit / risk
Low fidelity: sketches/paper	Brainstorming, conceptual layout, task flow and early user reactions. Quick to make and change.	A single screen hides progression; rough materials may not reveal visual detail, interaction feel or response time.
Medium fidelity: limited computer simulation	More engaging scenarios and subtler design issues, with selected behaviour represented.	Feedback may focus on tiny details. Users may hesitate to alter the design; managers may mistake it for a product.
High fidelity: closer appearance/interaction	Refining realistic interaction and presentation as the design develops.	Realism costs more; unimplemented features, error paths or backend work can still be missing.

Attributes of useful sketches

Quick, timely, disposable and plentiful. Invest in exploring the concept, not polishing the drawing. Make several alternatives, use a clear sketch vocabulary, and keep the rendering consistent with the actual maturity of the design. Sketches should provoke possibilities rather than merely confirm the first choice.

Pick the evidence the prototype can support

A paper screen can reveal an unclear label or unexpected grouping. It cannot demonstrate a production system's latency. A clickable mock-up may show navigation but still say little about real data, validation or automated recommendations. Report the representation and its limits when interpreting reactions.

Fidelity is not functional coverage

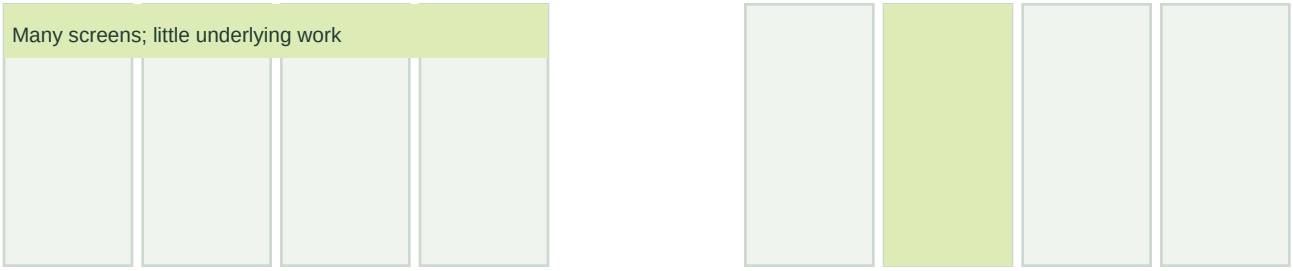
A polished set of screens can be broad but shallow. A rough prototype can explore one interaction in depth. Always specify both **how realistically it behaves/looks** and **which tasks/features it actually supports**.

PROTOTYPE SCOPE AND LIFECYCLE

Breadth, depth and what becomes the product

HORIZONTAL: BROAD SURFACE

VERTICAL: SELECTED DEPTH



Breadth is not the same as visual fidelity.

One/few features explored in depth

Scope	What is represented?	Example
Horizontal	A broad surface interface, with little or no underlying functional depth.	Show entry, history, goals and settings screens without a working service.
Vertical	A few selected functions explored in depth.	Implement the full correction of a meal quantity while other features remain absent.
Scenario	A fixed scripted use path, with little/no deviation supported.	Only log one specified breakfast and view its summary.

Prototype-to-product strategies

Strategy	What happens to the prototype?	Practical consequence
Throw-away	Used for reactions and learning, then discarded.	Keep construction rapid; findings inform the next design.
Incremental	Separate components are prototyped/tested, then integrated into the product.	Plan component boundaries and integration.
Evolutionary	The prototype itself changes repeatedly and becomes the product.	Do not confuse appearance of completion with readiness or completeness.

Two uses of the word scenario

In task modelling, a scenario is a concrete narrative about activity. In **scenario prototyping**, the emphasis is the limited fixed path implemented for that use. A convincing demonstration of one path does not show that alternative or error paths work.

Two-line model: choose a prototype for comparing overall navigation.

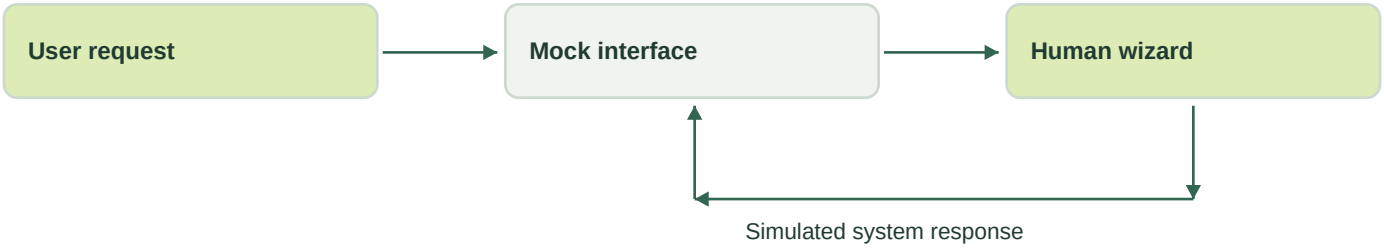
Use horizontal coverage to expose the main screens and relationships across features.
Keep backend depth minimal if the question concerns navigation rather than completed processing.

PROTOTYPE METHODS

Choose what to simulate and how

Method	How it works	What to keep in mind
Storyboarding	Key frames show interface states at important points in an interaction.	Shows progression; does not imply unrestricted interaction.
PICTIVE	Sticky notes, plastic overlays and office supplies form a collaborative, movable interface.	People demonstrate actions and revise pieces in the session; record only with appropriate consent.
Drawing packages	Computer-drawn scenes represent screens.	A thin horizontal representation; limited interaction feel.
Scripted simulation	Simple inputs trigger prepared transitions along a narrow task path.	Looks plausible within the script; deviation can break the simulation.
Interface builders	Lay out real standard widgets and selectively add functionality.	Concrete controls, but can constrain thinking or poorly fit rich graphical interaction.
Wizard of Oz	A human interprets input and supplies simulated system output.	Allows complex/unimplemented behaviour to be explored; does not prove the automation works.

Wizard of Oz: the response comes from a person



The wizard may be visible or hidden and follows a response process. For your proposed companion, a person could supply a meal suggestion through the mock interface. The study may reveal whether the interaction is understandable; it does not establish that an AI can produce equivalent suggestions reliably.

Choose tools by the question, not by polish

The lecture lists interface-building/prototyping tools as examples. Their availability can change. More important is whether the representation supports the target task, alternatives, feedback and realistic failure behaviour. A popular tool is not evidence that the design fits users.

Make limits explicit during feedback

Say which parts are simulated, which paths work, and what you want to learn. Ask people to react to concrete screens or interactions, rather than relying on a verbal sales description.

DESIGN PRINCIPLES

Simplify the task without hiding its meaning

Principle	What it means in practice	Example
Visibility	Make important actions and states discoverable.	Do not make a necessary action depend only on an undisclosed gesture.
Affordance / signifier	Support possible actions and communicate them through appropriate cues.	An input looks editable and has a clear label; a button communicates an action.
Constraints	Restrict invalid or unavailable actions.	Prevent choosing an unavailable date and explain the prerequisite.
Internal consistency	Similar operations behave alike within the application.	Save has the same meaning and outcome across screens.
External consistency	Follow relevant conventions across related products/devices.	Use familiar editing shortcuts and understandable controls.
Simplicity	Remove inessential complexity and use regular patterns.	Reduce redundant decoration while retaining needed instructions and feedback.
Learnability	Use familiar concepts to explain the unfamiliar.	A calendar metaphor supports date selection when its limits remain clear.

Simplicity techniques from the lecture

Remove inessential features/details: question every label, control, colour or decoration.
Use a regular pattern: consistent font, colour, line width, spacing and orientation.
Give an element double duty: combine compatible information or functions when their meaning stays clear.

Removing a necessary label may reduce the element count while increasing mental work. Test whether the design remains complete and understandable. A metaphor or minimalist style can conflict with other principles; judge the actual task consequences.

Two-line model: an action is available only through an unfamiliar swipe.

The gesture hides an important command, weakening visibility and first-use learnability.
Offer a discoverable control or clear cue that communicates the action and its effect.

The slides use "affordance" for graphical styling. The useful reading distinction remains: possible action versus the signifier that communicates it.

VISUAL DESIGN

Use layout, colour and type deliberately

Element	Lecture guidance	Reason / application
Grouping	Put related items together and avoid implying false relationships.	Keep a quantity label beside its own field; separate unrelated settings.
Hierarchy	Establish different levels of importance.	Use size, weight and selective colour to guide attention to the primary task.
Whitespace / alignment	Use space, margins and alignment to preserve simplicity.	Groups can be clear without a separate border around each item.
Grids	Use an organizing structure for layout and adaptation.	Align related content and maintain relationships as available space changes.
Colour quantity / saturation	Use few purposeful colours; avoid excessive saturated colour.	Unnecessary variation can make the design distracting and visually fatiguing.
Contrast	Maintain readable foreground/background separation.	A dark background does not automatically make all colours or text readable.
Typography	Use familiar readable fonts for long blocks; unusual styles may suit short salient text.	A logo and a page of instructions impose different reading demands.

Pre-attentive processing: make the signal stand out

Simple distinctive features can be noticed rapidly before detailed serial inspection. A regular visual background makes a meaningful difference more salient. If every control differs in colour, shape and type, the important exception may no longer stand out. The slide's roughly 200 ms comparison is an illustration, not a guaranteed detection time for every display.

Serif and sans serif are categories, not guarantees

Serif faces have small finishing strokes; sans serif faces lack them. The lecture discusses common visual associations, but readability still depends on the actual font, size, contrast, spacing, text length and user. Choose for the task rather than assuming one category is always superior.

Two-line model: improve a cluttered logging form.

Group related inputs with whitespace and alignment, preserving a clear label for each field.
Use readable type, strong contrast and selective emphasis so the main action is easy to find.

EVALUATION DURING DESIGN

Participatory design and analytical methods

Participatory design: users are collaborators

Representative **end users** take an active role during design, bringing expertise in their work and context. Managers or representatives alone cannot replace direct access to the people doing the work. Use concrete visuals and demonstrations so people can react and suggest changes throughout iteration.

Access can be difficult or expensive. Users are not automatically interface designers and may not specify the best solution from scratch. Their experiences inform the design; their first feature request is not an instruction to abandon investigation or alternatives.

Approach	Main question	Suitable evidence / limit
Participatory design	How can users' contextual knowledge shape the design?	Active collaboration and reactions; keep testing assumptions.
Heuristic evaluation	Where does the design violate useful usability rules?	Broad analytical inspection; evaluator bias and missed problems remain possible.
Cognitive walkthrough	Can a specified new user discover the task and understand the result?	Step-by-step analysis of exploratory learning; narrow and time-consuming.
Predictive models	How efficient is a specified action/movement likely to be?	Model estimates depend on the sequence, parameters and assumptions.
Empirical usability testing	What happens when representative users actually perform tasks?	Observed performance and experience; developed further in Lecture 06.

Analytical is not the same as observed

Heuristic inspection, cognitive walkthroughs and predictive models can evaluate without recruited users performing the task. They generate reasoned findings or predictions, not direct proof of how all real users will behave. Combine with empirical work when the question calls for it.

Two kinds of walkthrough

Your **project walkthrough** is a presentation and demonstration for audience feedback. A **cognitive walkthrough** is the structured analytical method using four questions at each task step. A presentation can use that method, but the terms do not automatically mean the same activity.

HEURISTIC EVALUATION

Nielsen's ten heuristics - with examples

Heuristic	Observable example / improvement
1. Visibility of system status	Show "Saving" then "Saved" with the relevant result.
2. Match between system and real world	Use terminology and ordering familiar from users' work.
3. User control and freedom	Offer Undo and a clear way out of an unwanted dialog.
4. Consistency and standards	Keep similar actions alike and follow appropriate conventions.
5. Error prevention	Reject an impossible quantity before committing the record.
6. Recognition rather than recall	Show recent foods instead of requiring an exact remembered code.
7. Flexibility and efficiency of use	Provide shortcuts for frequent users alongside discoverable menus.
8. Aesthetic and minimalist design	Remove irrelevant content that competes with task information.
9. Help users recognize, diagnose, recover from errors	Explain what went wrong, identify the affected input and state a recovery step.
10. Help and documentation	Provide findable, task-focused instructions when users need them.

Make findings specific

Inspect concrete task situations using the heuristics. Record the location/action, the violated rule, the likely consequence and a proposed change. One problem can relate to several heuristics; explain the strongest connection rather than attaching a list without reasoning.

Strengths and limits

Inspection avoids many participant-recruitment issues, but suitable experts may be difficult or costly to find. Application-domain knowledge helps. Evaluators can miss important problems, report many trivial ones or bring personal biases.

The graph on PDF p. 87 shows **diminishing returns**: more evaluators tend to find more problems, but the curve flattens. It is not a guarantee that a fixed number will identify every important issue.

Two-line model: distinguish error prevention from error recovery.

Prevention blocks or reduces an invalid action before its harmful consequence.
Recovery explains an error and helps users correct or reverse it after it occurs.

COGNITIVE WALKTHROUGH

Check discovery at each task step

Identify typical users and sample tasks. Step through the required actions, considering the user's knowledge at that point. The focus is **exploratory learning**, rather than maximum speed on a highly practised routine.

Question, in lecture order	Failure example	Targeted response
Q1. Will users be trying to produce this effect?	The task needs a correction, but users believe the existing record cannot be changed.	Make the editable record model and goal understandable.
Q2. Will the user notice the correct action is available?	An edit control exists only in a hidden gesture or off-screen menu.	Provide a noticeable edit route.
Q3. Will the user know the correct action is the right one?	A visible control has an ambiguous label, so its connection to correction is unclear.	Use a clear action label and context.
Q4. Will the user understand the feedback?	After editing, the record gives no clear indication of the saved quantity.	Show the updated value and meaningful confirmation.

Notice versus understand

A visible action can fail Q3 if its meaning is unclear. A user can understand an action before clicking yet fail Q4 because the result is missing or ambiguous. Do not assume that "a message appeared" proves the user noticed or understood it.

Record an argument, not just YES/NO

Document the user assumptions, where a problem may occur, why that assumption is plausible, side issues and possible changes; finish with a summary. A YES/NO label needs evidence about the particular user, task and interface state.

Relationship to Norman's action cycle

Both consider how goals become actions and results become understanding. The walkthrough has **four task-step questions**; Norman's model has **seven stages**. Do not substitute one ordered list for the other.

Choose the right method

For broad rule violations, inspect with heuristics. For first-use discoverability of a specific sequence, use a cognitive walkthrough. For actual first-use behaviour, observe representative users. Each answers a different question.

PREDICTIVE MODELS | GOMS AND KLM

Turn an action sequence into an estimate

GOMS element	Meaning	Text-deletion example
Goals	States the user wants to achieve.	Delete a word.
Operators	Elementary perceptual, motor or cognitive acts.	Point, select, press a key.
Methods	Procedures: sequences of operators that achieve goals.	Select the word, then press Delete.
Selection rules	Choose between alternative methods.	Use selection/Delete or repeated Backspace under specified conditions.

KLM: sum the time contributions of the operations used

T = nK tK + nP tP + nH tH + nD tD + nM tM + total response time. The lecture writes the sum of total category times as TK + TP + TH + TD + TM + TR. Count the actual operators before inserting values.

Operator	Lecture meaning	Use it when the sequence includes...
K	Keystroke	The specified key/button activation.
P	Point to a target with a mouse	Moving the pointer onto a target.
H	Home the hands on a device	Repositioning hands between relevant devices/positions.
D	Draw a straight line	A drawing operation.
M	Mental preparation	The stated mental preparation operation(s).
R	System response	The stated waiting/response intervals.

Worked practice sequence

Point to dropdown, activate it, point to an item, select it, point to Submit, activate it. For this exercise, treat each activation as K; use **3P + 3K + 1M + 1R**, with no H or D.
Given **P=1.1, K=0.2, M=1.35, R=0.02 seconds**:
T = 3(1.1) + 3(0.2) + 1.35 + 0.02 = 5.27 seconds.

These are explicit practice assumptions. Do not silently add a homing operation, extra mental preparation or response interval. Another task or timing convention may require different counts/values. Predictions do not establish observed accuracy, user satisfaction or first-use learnability.

PREDICTIVE MODELS | FITTS' LAW

Distance and target size shape pointing time

T = k log2(D/W + 1)
T: predicted pointing/movement time. **D:** distance to the target. **W:** target width along the movement direction. **k:** movement-time constant; the slide gives approximately 200 ms/bit.

The slide mixes **A/W** in its formula with **D/S** in its definitions. They describe the same roles here: A = D (distance) and W = S (target size). Use consistent notation and units. This guide follows the lecture's no-intercept formula; do not substitute another variant without stating it.

SAME DISTANCE: A WIDER TARGET IS EASIER TO POINT TO



Worked calculation and design consequences

Given **D=300 px, W=100 px, k=0.2 seconds/bit**:
D/W + 1 = 300/100 + 1 = 4.
T = 0.2 log2(4) = 0.2(2) = 0.4 seconds.
At fixed distance, increasing width reduces predicted time. At fixed width, reducing distance reduces time. Doubling both distance and width leaves D/W unchanged, so this formula predicts the same time.

This estimates aimed pointing, **not the click or whole task**. Use comparable units for D and W; convert 200 ms to 0.2 seconds if reporting seconds. If replacing a KLM pointing estimate with a Fitts estimate, do not count the same movement twice.

Final recall checklist

Models: concepts and relationships; metaphors transfer knowledge.
Prototypes: fidelity differs from breadth/depth; throw-away differs from evolutionary.
Design: make actions visible; preserve consistency; simplify without hiding meaning.
Evaluation: participatory collaboration; ten heuristics; four walkthrough questions.
Prediction: GOMS structures goals/actions; KLM sums operations; Fitts estimates pointing.

The lecture contains some older footer years and historical product/tool examples. Use the supplied file for the concepts; those examples are not current product recommendations or new assessment dates.

ACTIVE RECALL | SHORT ANSWERS

Two-line models 1-8

Recall before reading the answer. Keep both essential points; equivalent accurate wording can count.

1. Define a conceptual model and distinguish it from visual styling.

A conceptual model explains the system's concepts, operations and relationships at a high level.
Visual styling concerns their presentation, such as layout, colours and typography.

Lecture 05 | PDF pp. 3-5, 60

2. Why can a familiar metaphor help without being copied literally?

A metaphor transfers familiar knowledge to help users understand an unfamiliar interaction.
Keep useful correspondences, but avoid copying limitations or behaviours that conflict with the task.

Lecture 05 | PDF pp. 6-7, 58

3. Name the four interaction types listed in the lecture.

Instructing issues commands; conversing uses a conversational exchange.
Manipulating acts on objects; exploring moves through a virtual environment or physical space.

Lecture 05 | PDF p. 8

4. State the three defining characteristics of direct manipulation.

Continuously represent relevant objects and actions, and use physical actions instead of complex syntax.
Provide rapid, reversible actions with immediate feedback on the object of interest.

Lecture 05 | PDF p. 11

5. Why prototype early instead of implementing the first idea fully?

A prototype lets the team explore alternatives and obtain concrete user reactions at low cost.
Its low investment makes unsuitable ideas easier to discard or revise before implementation.

Lecture 05 | PDF pp. 21-25

6. Give one benefit and one limitation of a low-fidelity sketch.

A rough sketch is quick to change and directs discussion toward high-level concepts.
It may not reveal realistic interaction feel, visual detail or system response time.

Lecture 05 | PDF pp. 24-28

7. How does a storyboard differ from a single screen sketch?

A single sketch shows one appearance or interface state.
A storyboard links key frames to show progression through an interaction or scenario.

Lecture 05 | PDF pp. 25, 29-30

8. Distinguish horizontal, vertical and scenario prototypes.

Horizontal prototypes cover a broad interface surface; vertical prototypes explore selected functions in depth.
Scenario prototypes support a specific scripted use path, with little or no deviation.

Lecture 05 | PDF pp. 33, 36

ACTIVE RECALL | SHORT ANSWERS

Two-line models 9-16

Recall before reading the answer. Keep both essential points; equivalent accurate wording can count.

9. Why are fidelity and functional coverage different dimensions?

Fidelity concerns how closely the representation resembles the intended interaction or appearance.
Coverage concerns which features or task paths it supports, so polished screens can still be shallow.

Lecture 05 | PDF pp. 22-23, 32-33

10. Distinguish throw-away, incremental and evolutionary prototyping.

Throw-away prototypes inform design and are discarded; incremental work adds tested components to the product.
Evolutionary prototyping repeatedly refines the prototype itself until it becomes the product.

Lecture 05 | PDF p. 34

11. Explain PICTIVE and how users interact with it.

PICTIVE builds a collaborative interface using office supplies, sticky notes and plastic overlays.
People move the pieces to demonstrate interaction and modify the proposed design during the session.

Lecture 05 | PDF p. 31

12. Explain Wizard of Oz and one limit of its evidence.

A human wizard simulates responses behind a real or mock interface for unimplemented functionality.
User reactions can inform interaction design, but they do not prove equivalent automated functionality is feasible.

Lecture 05 | PDF pp. 43-45

13. Why can polished prototypes make feedback less useful?

People may focus on small details, hesitate to change the design or assume the prototype is finished.
Explain its limits and invite feedback on the workflow and alternatives, not just its appearance.

Lecture 05 | PDF p. 32

14. Distinguish internal from external consistency with examples.

Internal consistency keeps similar operations alike within an app, such as the same Save behaviour.
External consistency follows relevant conventions across apps or devices, such as familiar editing shortcuts.

Lecture 05 | PDF pp. 52-54

15. Explain simplicity without equating it with hiding every control.

Simplicity removes inessential elements and uses regular patterns to reduce processing effort.
Retain the labels, controls and feedback needed to make the task understandable and complete.

Lecture 05 | PDF pp. 56-57

16. Give two justified visual-design improvements for a crowded form.

Group related fields with whitespace and alignment so their relationships are easier to scan.
Use clear hierarchy, readable type and selective colour contrast to emphasize important actions.

Lecture 05 | PDF pp. 60-72

ACTIVE RECALL | SHORT ANSWERS

Two-line models 17-24

Recall before reading the answer. Keep both essential points; equivalent accurate wording can count.

17. What makes participatory design more than an end-of-project user test?

Representative end users actively collaborate throughout design as experts in their work context.
Their feedback informs repeated revisions; they are not expected to be expert interface designers.

Lecture 05 | PDF pp. 75-80

18. Distinguish heuristic evaluation from a cognitive walkthrough.

Heuristic evaluation inspects a design against usability rules across a broad range of use.
A cognitive walkthrough follows a task step by step to assess a new user's exploratory learning.

Lecture 05 | PDF pp. 81, 85, 89-90

19. List the four cognitive-walkthrough questions in order.

Will users try to produce the effect, and notice that the correct action is available?
Will they know the action is right, and understand the feedback after performing it?

Lecture 05 | PDF p. 90

20. A user notices Submit but cannot tell whether it completes the task. Diagnose the walkthrough issue.

The user notices the action but cannot connect its meaning to the intended effect: question 3.
Use a clear action label and supporting context that communicate what submission will accomplish.

Lecture 05 | PDF p. 90

21. Explain GOMS using a text-deletion task.

Goal: delete a word; operators: pointing, selection and keystrokes; method: select the word then delete.
Selection rules choose between that method and alternatives such as repeated Backspace.

Lecture 05 | PDF p. 93

22. Name the six KLM operators listed in the lecture.

K: keystroke; P: pointing; H: homing the hands on a device.
D: drawing a straight line; M: mental preparation; R: system response.

Lecture 05 | PDF p. 94

23. Calculate 3P + 3K + M + R using P=1.1, K=0.2, M=1.35 and R=0.02 seconds.

T = 3(1.1) + 3(0.2) + 1.35 + 0.02.
T = 5.27 seconds for exactly the stated operators and supplied practice timings.

Lecture 05 | PDF p. 94

24. State the lecture's Fitts formula and its design implication.

$T = k \log_2(D/W + 1)$, with D as target distance and W as width along the movement direction.
Larger targets or shorter distances reduce predicted pointing time when other terms remain fixed.

Lecture 05 | PDF p. 95

ACTIVE RECALL | MULTIPLE CHOICE

MCQ practice 1-8

Choose one best answer. Explain the concept before checking the answer key.

1. What does a conceptual model primarily describe?

A. Only its final colours and fonts
B. Its complete source code and database implementation
C. The number of participants in a usability experiment
D. The system's high-level concepts, organization and relationships

2. Why challenge the claim that everyone will happily wear special glasses for a new TV?

A. Technological novelty guarantees acceptance
B. Users must adapt to every design choice
C. It is an assumption about users that needs evidence
D. A conceptual model removes the need for research

3. A shopping cart helps users understand collecting items before purchase. What is it doing?

A. Proving that all physical cart properties must be copied
B. Using a familiar metaphor to explain an unfamiliar interaction
C. Measuring pointing time
D. Replacing every need for feedback

4. Which example best fits the lecture's exploring interaction type?

A. Moving through a virtual museum to discover exhibits
B. Typing a command to delete a file
C. Filling a fixed postal-address form
D. Calculating the mean of task times

5. Which set best characterizes direct manipulation?

A. Hidden commands, complex syntax and delayed results
B. A compulsory natural-language conversation
C. A static image with no representation of objects
D. Visible objects, physical actions and rapid reversible changes with immediate feedback

6. Why can menus and forms be easier to learn than an unfamiliar command language?

A. They remove all possible errors
B. They always outperform commands for every expert
C. They present choices for recognition instead of requiring exact syntax recall
D. They never require labels

7. What is a discoverability risk of speech dialog?

A. All spoken requests are necessarily understood
B. Users may not know what commands or requests the system supports
C. Speech always displays a complete menu
D. Natural language guarantees a correct mental model

8. Why build a rough prototype early?

A. Explore alternatives and obtain reactions before investing heavily in implementation
B. Prove that the final system is complete
C. Avoid involving users
D. Guarantee accurate production response times

ACTIVE RECALL | MULTIPLE CHOICE

MCQ practice 9-16

Choose one best answer. Explain the concept before checking the answer key.

9. Which claim is least justified by testing a paper sketch?

A. Users can comment on the proposed grouping
B. The team can compare alternative conceptual arrangements
C. Participants can suggest a clearer action label
D. The finished system will respond within 100 milliseconds

10. What do a series of storyboard frames add to a single sketch?

A. A guarantee that the backend works
B. A population-level estimate of error rates
C. A representation of how the interaction progresses over time
D. Automatic support for every possible task branch

11. What best describes PICTIVE in the lecture?

A. A complete production database
B. Collaborative interface construction with sticky notes and plastic overlays
C. A statistical test for comparing means
D. A pointing-time equation

12. A polished prototype triggers discussion only about tiny visual details, while major workflow problems go unchallenged. What risk is illustrated?

A. Fidelity can narrow feedback and make people reluctant to challenge the design
B. Polish guarantees complete functionality
C. Only paper prototypes can have visual issues
D. More detail always produces broader feedback

13. A mock-up shows all major screens but has no working backend. Which coverage is emphasized?

A. Vertical depth across every function
B. A complete evolutionary product
C. A matched-participants design
D. Horizontal coverage

14. Only checkout is implemented in depth while most other functions are absent. Which prototype fits?

A. Purely horizontal prototype
B. A questionnaire
C. Vertical prototype
D. A persona

15. A prototype supports only one scripted path and fails if users deviate. What kind of limitation is this?

A. Complete task coverage
B. Scenario-based fixed-path coverage
C. A field trial with unrestricted use
D. Evidence that all error paths work

16. A prototype is discarded after collecting design feedback. Which strategy is this?

A. Throw-away prototyping
B. Evolutionary prototyping
C. Incremental integration
D. Selective coding

ACTIVE RECALL | MULTIPLE CHOICE

MCQ practice 17-24

Choose one best answer. Explain the concept before checking the answer key.

17. A prototype is repeatedly refined until it becomes the final product. Which strategy is this?

A. Throw-away prototyping
B. A fixed storyboard by definition
C. Between-subjects testing
D. Evolutionary prototyping

18. What is a trade-off of interface-builder tools?

A. They automatically implement every backend function
B. They cannot represent any interface controls
C. They offer real widgets but can constrain thinking to standard controls
D. They make user evaluation unnecessary

19. A human interprets requests and generates responses behind a mock interface. Which technique is this?

A. A Latin square
B. Wizard of Oz
C. Axial coding
D. An interval scale

20. Why does successful Wizard-of-Oz interaction not prove the proposed AI is technically feasible?

A. The observed responses may have been supplied by the human wizard
B. The prototype necessarily used a fully working AI
C. Users cannot react to simulated behaviour
D. Human simulation removes every design assumption

21. A crucial feature is accessible only through an undisclosed swipe. Which concern is most direct?

A. The ratio scale of the task time
B. The number of grounded-theory codes
C. The choice of participant matching variable
D. Visibility of available actions

22. Save has the same label and behaviour on every screen of one app. Which consistency is illustrated?

A. External consistency only
B. Random assignment
C. Internal consistency
D. Cognitive tracing

23. Which redesign best supports simplicity without losing required functionality?

A. Hide every label, including unfamiliar controls
B. Remove redundant decoration and group essential controls with consistent spacing
C. Delete a required confirmation solely to reduce the element count
D. Give every control a different colour and font

24. What is a good layout response to crowded, unrelated controls?

A. Group related items with whitespace and alignment, and separate unrelated groups
B. Draw a heavy border around every item regardless of its role
C. Remove all hierarchy
D. Use equal emphasis for every element

ACTIVE RECALL | MULTIPLE CHOICE

MCQ practice 25-32

Choose one best answer. Explain the concept before checking the answer key.

25. Why limit unnecessary colour and typography variations?

A. Every interface must use exactly one colour
B. Colour always replaces text successfully
C. Pre-attentive processing guarantees users notice every warning
D. A consistent background pattern lets meaningful differences stand out

26. Which typography choice best follows the lecture?

A. Use decorative display text for every paragraph
B. Assume every sans-serif font is equally legible
C. Use familiar, readable type for long text and reserve unusual styles for short salient uses
D. Choose type only by the newest fashion

27. What makes participatory design different from merely testing users at the end?

A. Managers always substitute for actual users
B. Representative end users actively collaborate in design decisions throughout iteration
C. Users must create every design from scratch
D. The first proposed design cannot be revised

28. Experts inspect an interface using Nielsen's ten heuristics. What is this method?

A. Heuristic evaluation
B. An empirical usability test by definition
C. A SUS questionnaire
D. A matched-pairs experiment

29. After a failed payment, a message identifies the problem and explains how to correct it. Which heuristic fits best?

A. Aesthetic and minimalist design only
B. Flexibility and efficiency of use only
C. Match between system and real world only
D. Help users recognize, diagnose and recover from errors

30. A frequent user can use keyboard shortcuts while novices can still use menus. Which heuristic fits?

A. Help and documentation only
B. Error diagnosis only
C. Flexibility and efficiency of use
D. System status only

31. A design lets users undo an unintended action and leave an unwanted dialog. Which heuristic is most direct?

A. A requirement for no feedback
B. User control and freedom
C. A ban on all constraints
D. The use of a nominal scale

32. What does the lecture's evaluator-number graph suggest?

A. Additional evaluators find more problems, with diminishing returns
B. One evaluator finds every important problem
C. Each additional evaluator always adds exactly the same number of findings
D. Expert inspection proves that no user testing is needed

ACTIVE RECALL | MULTIPLE CHOICE

MCQ practice 33-40

Choose one best answer. Explain the concept before checking the answer key.

33. What is the main focus of a cognitive walkthrough?

A. Maximum speed after years of expert practice
B. The production database's storage capacity
C. Which statistical test has the largest sample
D. Whether a new user can discover the action sequence and understand its results

34. In a walkthrough, the needed button is visible but its label does not suggest the desired outcome. Which question is most directly at risk?

A. Will the user notice that the correct action is available? only
B. What is the standard deviation?
C. Will the user know the correct action is the right one?
D. Which component will become the final product?

35. A new user clicks Save but cannot tell whether the item was stored. Which walkthrough question fails?

A. Will the user try to produce the intended effect? only
B. Will the user understand the feedback?
C. Is the prototype horizontal?
D. Was the user randomly assigned?

36. What do the letters in GOMS represent?

A. Goals, Operators, Methods and Selection rules
B. Graphics, Objects, Menus and Screens
C. Groups, Outcomes, Means and Samples
D. Goals, Observations, Medians and Scales

37. In GOMS, choosing between selecting a word then pressing Delete and repeated Backspace is described by:

A. A SUS conversion rule
B. A colour-contrast ratio
C. A task-success percentage
D. Selection rules for alternative methods

38. For this practice sequence, count 3P + 3K + 1M + 1R. Given P=1.1, K=0.2, M=1.35 and R=0.02 seconds, what is the KLM total?

A. 2.67 seconds
B. 4.52 seconds
C. 5.27 seconds
D. 6.37 seconds

39. Using T = k log2(D/W + 1), with D=300 px, W=100 px and k=0.2 seconds/bit, what is the predicted pointing time?

A. 0.6 seconds
B. 0.4 seconds
C. 0.2 seconds
D. 4 seconds

40. What does the lecture's Fitts model predict if both target distance and width double, with k unchanged?

A. The same pointing time because D/W is unchanged
B. Twice the pointing time
C. Half the pointing time
D. Zero pointing time

CHECK YOUR REASONING

MCQ answers 1-20

1. D - The conceptual model explains how the system is organized and operates, without requiring implementation detail.

Lecture 05 · PDF pp. 3–5

2. C - Conceptualization should expose assumptions and prevent baseless claims about user acceptance.

Lecture 05 · PDF pp. 3–4

3. B - Metaphors transfer useful familiar knowledge, but software need not copy all properties of the physical example.

Lecture 05 · PDF pp. 5–7, 58

4. A - Exploring involves moving through a virtual environment or physical space.

Lecture 05 · PDF p. 8

5. D - The lecture lists continuous representation, physical actions and rapid reversible actions with immediate feedback.

Lecture 05 · PDF p. 11

6. C - Visible choices can reduce recall demands; command languages may remain useful for expert work.

Lecture 05 · PDF pp. 10–12

7. B - Speech can be approachable yet leave capabilities and valid requests unclear.

Lecture 05 · PDF p. 13

8. A - Cheap, changeable representations make alternatives easier to evaluate and discard.

Lecture 05 · PDF pp. 21–25

9. D - A paper representation does not reproduce the implemented system’s response time or interaction feel.

Lecture 05 · PDF pp. 25, 28

10. C - A storyboard uses key frames to make important moments and progression visible.

Lecture 05 · PDF pp. 25, 29–30

11. B - Movable office supplies let people demonstrate interactions and modify designs during the session.

Lecture 05 · PDF p. 31

12. A - The lecture warns about reactions in the small, reluctance to change the design and mistaking the prototype for a product.

Lecture 05 · PDF p. 32

13. D - A horizontal prototype represents broad surface functionality without deep underlying implementation.

Lecture 05 · PDF p. 33

14. C - A vertical prototype explores a few selected features in depth.

Lecture 05 · PDF p. 33

15. B - Scenario prototypes and scripted simulations support selected fixed uses rather than unrestricted exploration.

Lecture 05 · PDF pp. 33, 36

16. A - A throw-away prototype exists to inform design, rather than becoming the final implementation.

Lecture 05 · PDF p. 34

17. D - Evolutionary prototyping develops the prototype itself into the product; incremental development adds separately developed modules.

Lecture 05 · PDF p. 34

18. C - Builders support concrete controls and broad mock-ups, while potentially limiting novel or rich graphical interaction.

Lecture 05 · PDF pp. 41–42

19. B - A human wizard simulates system responses, making unimplemented or complex functionality testable.

Lecture 05 · PDF pp. 43–45

20. A - The technique tests an interaction concept without demonstrating that an automated implementation can deliver the same responses.

Lecture 05 · PDF pp. 43–45

CHECK YOUR REASONING

MCQ answers 21-40

21. D - Gestures can act as hidden commands; a usable design needs discoverable routes to important functions.

Lecture 05 · PDF pp. 48–49

22. C - Internal consistency applies across operations within an application; external consistency concerns other products/devices.

Lecture 05 · PDF pp. 52–54

23. B - Simplicity removes inessential complexity while retaining the information and controls needed for the task.

Lecture 05 · PDF pp. 56–57, 61–65

24. A - Grouping, hierarchy, whitespace, alignment and grids can clarify structure without extra clutter.

Lecture 05 · PDF pp. 61–65

25. D - Excessive variation competes for attention; selective distinction can make relevant information easier to find.

Lecture 05 · PDF pp. 66–69

26. C - The lecture distinguishes long reading from short branding/salient text; actual readability still depends on presentation.

Lecture 05 · PDF pp. 70–72

27. B - Users are first-class collaborators with contextual expertise, rather than only passive test participants.

Lecture 05 · PDF pp. 75–80

28. A - Heuristic evaluation is an analytical inspection guided by rules of thumb; it does not require recruited users performing the tasks.

Lecture 05 · PDF pp. 81–88

29. D - The message explains an error and a recovery action; this differs from preventing the invalid action before it occurs.

Lecture 05 · PDF p. 85

30. C - Accelerators can support experienced users without forcing novices to memorize commands.

Lecture 05 · PDF p. 85

31. B - Undo and escape support control and freedom when users take an unwanted route.

Lecture 05 · PDF p. 85

32. A - The curve rises but flattens; it does not promise universal detection rates or eliminate evaluator bias.

Lecture 05 · PDF pp. 87–88

33. D - Walkthroughs assess exploratory learning for specified users and tasks, not primarily highly practised performance.

Lecture 05 · PDF pp. 81, 89–90

34. C - Noticing a control is different from understanding that it leads to the desired effect.

Lecture 05 · PDF p. 90

35. B - The breakdown concerns interpreting the system's response after the action.

Lecture 05 · PDF p. 90

36. A - GOMS separates intended goals, elementary actions, procedures and rules for choosing among procedures.

Lecture 05 · PDF p. 93

37. D - Methods are action procedures; selection rules determine which method is used when alternatives exist.

Lecture 05 · PDF p. 93

38. C - T = 3(1.1) + 3(0.2) + 1.35 + 0.02 = 5.27 seconds. These are supplied practice constants, not a timing table in the lecture.

Lecture 05 · PDF p. 94

39. B - D/W + 1 = 4; log2(4) = 2; T = 0.2 × 2 = 0.4 seconds. This predicts pointing, not the entire task.

Lecture 05 · PDF p. 95

40. A - In this model difficulty depends on the distance-to-width ratio; increasing both equally preserves that ratio.

Lecture 05 · PDF p. 95